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SmartCHANGE

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SmartCHANGE: AI-based long-term health risk evaluation for driving behaviour change strategies in children and youth

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***Abstract*—The SmartCHANGE project, a Horizon-Europe Research & Innovation project that has been ongoing since May 2023 and is scheduled to conclude in April 2027, aims to develop AI-driven decision-support tools to identify and reduce the long-term risk of non-communicable chronic diseases (NCDs) in children and youth. NCDs have become a significant public health burden in developed countries, with common risk factors including obesity, low physical fitness, and unhealthy lifestyle habits. Childhood and adolescence are critical periods for establishing healthy habits, and early intervention can prevent or delay the onset of NCDs later in life. However, current tools for identifying high-risk individuals mainly focus on adults, leading to missed opportunities for early detection. The SmartCHANGE project aims to address this gap by creating trustworthy AI tools that accurately assess risk factors in children and youth and promote optimised strategies for risk reduction. The paper outlines the plan for the project from the project's proposed methodologies, including dataset utilisation, data cleaning, risk prediction models (basic and complex), and the implementation of federated learning. Additionally, it discusses user experience aspects such as participatory design, suggested applications, and the evaluation of project outcomes.**

***Index Terms*—SmartCHANGE, non-communicable chronic diseases, trustworthy AI, decision-support tools, risk prediction**

I. Introduction

In developed countries, non-communicable chronic diseases (NCDs) have become the leading public health burden over the past few decades. According to the World Health Organization (WHO), more than 70% of mortality in the European region can be attributed to NCDs, [1]. Many NCDs have common risk factors, including obesity, low levels of physical fitness, and unhealthy lifestyle habits such as insufficient physical activity, sedentary behaviour, poor nutrition, inadequate sleep duration, smoking, and excessive alcohol consumption. These risk factors often occur together and can interact to increase the risk of NCDs. Adopting a healthy lifestyle can enhance physical, social, and mental well-being, particularly among young individuals, while minimising the risks of morbidity and mortality associated with NCDs, [2], [3], [4].

Clinical prevention strategies for NCDs have usually focused on adults, as biological risk factors typically manifest during adulthood. However, recent evidence suggests that targeting interventions toward children and adolescents can be a more effective approach to reducing the risk of NCDs through behaviour modification, [5]. There are several reasons for this recommendation. Firstly, although NCDs may not be apparent during childhood and adolescence, early indicators of these diseases can already be present. By addressing risk factors and promoting healthy behaviours during the development period, it is possible to prevent or delay the occurrence of NCDs later in life, [6]. Secondly, childhood and adolescence are crucial stages for establishing healthy lifestyle habits. Since risk factors for NCDs often persist from childhood into adulthood, [7], the risk assessment should begin already in childhood, and a reduction in risk-factor levels between childhood and adulthood may potentially lower the incidence of NCDs. Lastly, NCDs in adolescents contribute significantly to global health challenges, as nearly one in five adolescents worldwide are overweight or obese, and approximately one in seven experiences a mental disorder, [8].

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Identifying high-risk individuals for future health problems is crucial for targeted preventive interventions. As stated above, current tools mainly focus on adults, [9], thus missing the opportunity to identify high-risk individuals during childhood and adolescence, where lifestyle habits are formed. Identifying children and youth at risk from future NCDs in routine clinical practice is proving challenging with such tools. For example, only 35% of doctors in the UK know that recommendations for physical activity exist, and only 13% were able to state the recommended duration per week. Over 80% of parents of inactive children wrongly consider their children to be sufficiently active, [10]. Developing risk prediction tools for children and youth would thus significantly improve the prevention of NCDs and promote cost-effective approaches.

Most health-risk prediction models are based on conventional statistical methods, although machine learning is gaining popularity, [11], [12]. While both model types generally demonstrate comparable accuracy, machine learning has been suggested to have a slight advantage in some studies, [13]. To overcome the challenges of heterogeneous datasets that combine biomarkers, fitness data, and data from wearables, the project proposes a unified deep-learning approach that incorporates all relevant variables into one network. Despite being underutilised in risk prediction, deep neural networks offer greater flexibility in constructing and training models. The potential of deep networks has been demonstrated in recent work on chronic kidney disease, where even a relatively conventional architecture outperformed classical machine learning techniques, [14]. By leveraging deep supervision, an architecture and training method from computer vision that utilises intermediate concepts to deeply supervise hidden layers of a deep neural network, [15], the project will aim to achieve beyond-state-of-the-art performance. Furthermore, the project will draw upon transfer learning methods, [16], to effectively combine datasets, incrementally training only relevant network parts without compromising the rest.

The construction and evaluation of risk prediction models using machine learning can be prone to bias and lack of trust, [12], [17], necessitating correction before they can replace traditional statistical methods. A cutting-edge technique for privacy-aware machine learning is federated learning, combined with differential privacy, which enables collaborations across institutions, safeguarding patient privacy and confidentiality, [18]. The project must address issues like data representation, concept drift, and uncertainty balancing to build robust risk prediction models instead of solely optimising accuracy. Combining federated learning with explainable AI will be a significant advancement beyond the current state of the art.

Moving from the above considerations, the main objective of the SmartCHANGE project is to create trustworthy AI-

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driven decision-support tools that will help healthcare professionals and individuals identify risks and reduce the long-term risk of NCDs. These tools will try to accurately assess the risk factors in children and youth, including those that are difficult to detect, and promote the implementation of optimised strategies to lower the risk. In this paper, the main ideas of the SmartCHANGE project will be presented, [19]. Section II begins by outlining the datasets employed. Next, it describes the methods used for data cleaning, followed by the proposed techniques for risk prediction, which include initial basic prediction models, more complex models, and the implementation of federated learning. In Section III, the emphasis of the paper shifts to user experience. It starts with a discussion on participatory design, provides an overview of the suggested applications, and concludes with an evaluation of the project's outcomes.

II. Methodology

A. Datasets

The consortium of the SmartCHANGE project is conducting a survey of existing data and has so far identified 15 relevant datasets comprising approximately 200,000 subjects: SLOfit, Cardiovascular Risk in Young Finns (YFS), Amsterdam Born Children and their Development (ABCD), ACDSi, ICAD, UK Biobank, HELENA, PREFIT, ActiveBrains, EFIGRO, ACTIBATE, PACO, Fit-Aging, UP-DOWN, and DADOS. The main criteria for inclusion were that the data include longitudinal lifestyle data from childhood to adulthood (causality) or big cross-sectional lifestyle data. While some datasets are publicly available, others are exclusively accessible to consortium members, providing a distinct advantage for conducting the SmartCHANGE project. In particular, two cohort datasets stand out. SLOfit is a national monitoring system in Slovenia that has collected comprehensive data on children and youth's physical and motor development since 1982. The dataset covers all Slovenian schoolchildren and spans 13 consecutive years during their schooling period (6-19 years of age), making it one of the most comprehensive fitness datasets in the world. The Cardiovascular Risk in Young Finns Study began in 1980 and has followed the same individuals for 42 years to investigate how childhood lifestyle, biological factors, and psychological measures contribute to the risk of cardiovascular diseases in adulthood. Figure 1 illustrates the categories of data included in some of the available datasets (not all datasets are represented). Although ample data is presently encompassed in the datasets, it is imperative to declare that the consortium will not cease at these datasets. Instead, a systematic exploration for supplementary open data conforming to the inclusion criteria will be undertaken.

B. Data cleaning

Data cleaning methods will include both a manual and an automated approach. In the context of the manual approach, the interacting stakeholders willing to perform data cleaning can load their datasets and choose from a pre-defined list of cleaning actions. These actions will include the ability to remove duplicate entries, detect outliers, and remove noisy and irrelevant data. In the context of manual data cleaning, the Data Cleaning methods will include predicting inaccurate or erroneous data. Using the mean or the most common value for missing values in this context will be regarded in certain circumstances as the primary strategy; however, the entire data entry will be discarded for more significant features. For the abovementioned, the findings of previous research work, [20], will be exploited. Concerning the automated data-cleaning approach, depending on similar past stakeholders' (e.g., healthcare practitioners, physicians) data-cleaning actions on specific warnings (e.g., healthcare practitioners usually remove attributes containing nominal values, or physicians are usually interested in depicting the ranges of an attribute instead of unique instances), the same pattern-based cleaning actions will be applied. The cleaned datasets will be automatically generated without any user interaction. The latter will exploit the previous research outcomes, [20].

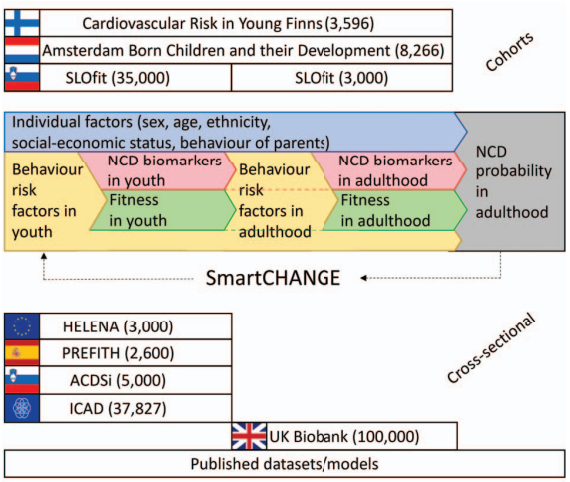


Fig. 1. Datasets used in the project.

In both cases, all the cleaning actions will be traced and recorded in appropriate log files, exploiting the previously developed methodology, [21], to track all the changes and keep a version history of the data cleaning process for potential rollbacks to older versions of the datasets. In order to improve and facilitate the cleaned datasets' reusability for further exploitation, datasets will be translated into a common syntactic and semantic representation as soon as they are considered error-free, which will be done following the guidelines of a specific SmartCHANGE HL7 Fast Healthcare Interoperability Resources (FHIR) data model that will have been primarily created based on previous work, [22]. The HL7 FHIR data model is essential for interoperability, standardisation, flexibility, patient-centricity, and developer-friendly healthcare data exchange, contributing to improved healthcare outcomes and delivery. It can facilitate interoperability, as FHIR provides a standardised approach to exchanging healthcare data among

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different organisations and systems. Moreover, its data model allows for seamless integration and interoperability, enabling Healthcare Information Systems (HIS) to share information and communicate effectively. In SmartCHANGE, this HL7 FHIR data model will contain a list of different HL7 FHIR Resources (i.e., discrete units of data representing a specific healthcare concept (e.g., Patients, Medication)) and their corresponding attributes that will be considered of high importance for SmartCHANGE's overall goal, having been derived from the SmartCHANGE ingested datasets and data sources.

C. Baseline predictive models for NCDs

The models developed in the project will relate to lifestyle behaviour such as physical activity, fitness, and biomarkers for NCDs such as BMI and blood lipids, and the lifetime risk of CVD and metabolic disease from childhood/youth to adulthood. Currently, the main challenge is that datasets and models relating to all these variables do not exist, as they would require tracking individuals from childhood/youth to old age when NCDs become common.

The first step to tackle this challenge is to build multiple simple models that predict one of these variables from other (predicted) variables and then connect them. For example, one model may predict fitness in young adulthood from fitness in childhood based on the SLOfit data. Then another model may predict blood biomarkers from fitness in adulthood. The third one may use these biomarkers and fitness information to predict NCDs. These models can be built with conventional machine-learning methods. Some preliminary research of this type was already performed by some of the partners involved in the SmartCHANGE project, [23], and one of our objectives is to build upon this preliminary research.

D. Unified predictive model on heterogeneous data for NCDs

The ambition of the SmartCHANGE project is to develop a methodology for building one unified model that spans all of the variables across multiple ages instead of building several simple models and combining them as described in the previous section. We plan to train such a model on all the available datasets. Further, we expect that incorporating all the data and the relations between them in one model will result in higher accuracy. When the prediction is made for a new person, the model is expected to make it regardless of which variables are available for this person, as long as they represent a large enough subset of the variables in the training data.

The unified model will be a neural network with a novel architecture to be developed in the project. The key advantage of neural networks is that layers can train them. Inputs can be provided for the earliest variables in time (early childhood) and causal chain (behaviours, which are the cause of NCD biomarkers, fitness and eventually NCDs as shown in Figure 1), or later ones. Similarly, the network can be supervised at different points, as Figure 2 illustrates. Another possibility is to use deep supervision, [15], where ground truth for both the final target variable for intermediate concepts can be utilised in training.

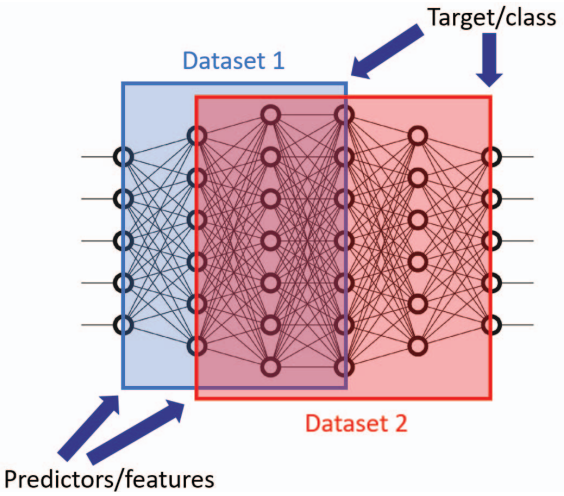


Fig. 2. Training of a neural network by layers depending on the dataset.

E. Federated learning

In federated learning, [24], multiple entities (clients) collaborate in solving a machine-learning problem under the coordination of a central server. Each client's data is stored locally and not shared; instead, the client trains a local model and only shares the encrypted weights of the trained model. The weights are then anonymously aggregated by the server into a general model, which is returned to the clients to be used for prediction. All these steps build up to one cycle of federated learning, illustrated in Figure 3.

Using federated learning in SmartCHANGE will enable building risk prediction models without any datasets or live data sources leaving a particular site. The methods will be built on top of baseline and unified models described in previous subsections using federated platforms, [25], [26]. We will also investigate federated analytics, i.e., federated outlier detection, data visualisations, and statistical tests that do not require data centralisation. The federated machine learning and analytics will be expanded with modules for federated explainable AI, e.g., based on counterfactual explanations, [27]. Ultimately, the federated approach will allow for the personalisation of risk predictions and corresponding explanations without sharing personal data, enabling privacy awareness and explainability, which are crucial for international AI projects (like SmartCHANGE) applied in healthcare.

F. Explainable models and visualization

Achieving explainability in predictive models can be accomplished through transparent and intricately explainable models or by employing post-hoc explanations such as feature relevance analysis, visual explanations, or model approximations. In the SmartCHANGE project, an advanced approach will

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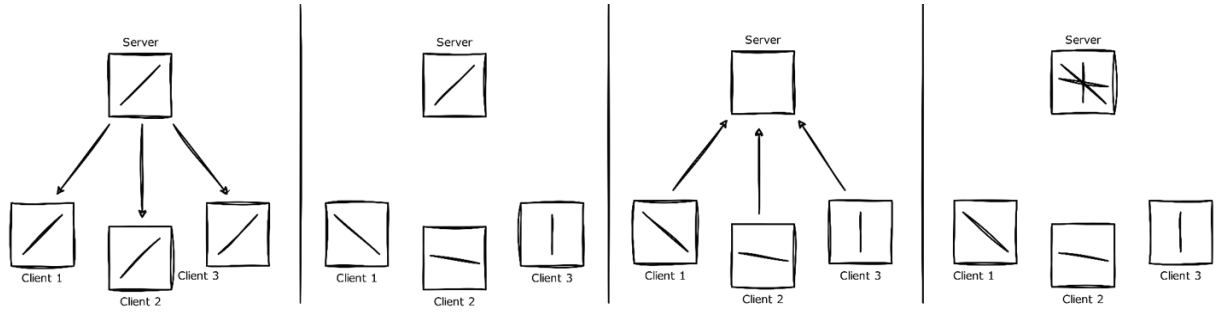


Fig. 3. Scheme of the federated learning cycle.

be employed, combining explainable models with the generation of counterfactual explanations for each patient, [26]. The counterfactual explanations are generated for each model prediction, aiming to simulate hypothetical scenarios and their associated future conditions, thereby enhancing interpretability and trustworthiness. The primary focus of the counterfactual generator in SmartCHANGE is to identify the most plausible and minimal lifestyle changes that can effectively reduce the risk of developing NCDs in the future. Consequently, in collaboration with medical experts, this approach facilitates the suggestion of intervention strategies prioritised according to the extent of required behavioural modifications.

The human-in-the-loop aspects will be addressed through visual analytics for self-auditing, model, and prediction understanding, [28]. Integration into application tools will be based on understanding by machine-learning experts and health professionals. The initial focus will be on providing solutions for instance-based explanations. Contextual visualisation of counterfactuals and interactive features will enable the evaluation of what-if scenarios and the utilisation of users' domain knowledge. Accuracy and uncertainty estimation regions will be incorporated. Overcoming the challenge of missing and heterogeneous data to prevent user bias represents a key advancement beyond the current state of the art. Achieving low latencies for optimal user experience will be addressed through solutions like Progressive Visual Analytics, [29]. Evaluation will adhere to standard visualisation design practices.

G. Measures to increase the robustness of the models

In order to enhance the robustness of the models, various measures will be implemented. The first measure to enhance model robustness is training data curation by employing various techniques to characterise structural changes and drifts in the data, which includes detecting out-of-distribution samples and analysing disparities between the training data and the target population. Preprocessing will further ensure that the training data adequately represents the target population.

The reliability and robustness of the models will be improved by addressing changes in the target population over time. Solutions such as concept drift, [30], and automatic detection of out-of-distribution samples will be applied. To tackle the challenge of robustness to missing and corrupt data, calibrated adversarial training, [31], will be employed. Instead of solely optimising for accuracy, the models will be tuned to balance generalisation accuracy and prediction uncertainty.

Modelling uncertainty is another key aspect of increasing the robustness of the models. The AI solution will be allowed to express uncertainty by acknowledging when it is uncertain or does not have sufficient knowledge to provide a definite answer. Expertise in probabilistic machine learning will be utilised to estimate the uncertainty of predictors. Furthermore, predictor performance profiling will be conducted to identify subgroups in the data where the model exhibits unusual inaccuracies, which will involve employing expertise in exceptional model mining techniques.

By implementing these measures, the models will be more robust, capable of adapting to changes in the target population and better equipped to handle missing and corrupt data. Additionally, the models will provide estimates of uncertainty and identify areas where inaccuracies may occur, enabling more reliable and informed decision-making. The available risk prediction tools mostly rely on biological factors and require information from biochemical essays, rendering them unsuitable for children and youth. Moreover, even when data on lifestyle behaviours are included in the risk prediction models, they are very generic and lack the necessary level of detail. The project aims to overcome these limitations by leveraging long-term cohort data to estimate the lifetime risk of CVD and metabolic disease from behavioural and biological risk factors measured in childhood and youth. Childhood and youth are the time of lifestyle formation, and hence, an optimal period to start with prevention strategies, [32]. A tool will be provided by SmartCHANGE that integrates all relevant behaviours and biomarkers and produces a consolidated risk score for future cardiovascular or metabolic disease.

To make it actionable, contributions of individual risk factors to aggregated risk will also be estimated, and guidance on personalised risk-lowering strategies will be offered, thus raising awareness about the importance of a healthy lifestyle in citizens and medical professionals.

III. Implementation

In order to attain practical impact, accurate risk-prediction models are required, and these models are embedded into

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tools that health practitioners and other relevant stakeholders can utilise to enhance the well-being of children and adolescents. The SmartCHANGE tools are set to encompass a web application aimed at healthcare professionals and mobile applications for children and their parents. These applications will be fashioned through a participatory design approach and appraised through proof-of-concept studies.

A. Participatory Design

Participatory design in a study involving children is crucial to understanding children's needs from their perspective, involving their caregivers and professionals, and ensuring that the research considers their unique experiences and requirements. The methodology in the SmartCHANGE project plans to include youth participatory action research, [33], to collect perspectives of young people on their risk behaviours and suitable behaviour change strategies. Moreover, it will involve SmartCHANGE users as co-researchers. To generate rich user insights and sustainable user scenarios, a novel combination of methods from participatory action research (concept mapping, [34], photo voice, [35], storytelling, [36]), with participatory methods from the field of product-system innovation (contextual inquiry, context mapping, co-design), [37], will be used. After completing the protocol, a workshop will be organised to train research teams facilitating this process at study sites.

B. Co-creation of trustworthy explainable AI predictions and models

Trustworthy explainable AI (XAI) is essential in predictive models, especially when working with sensitive data (children and young adults). Therefore, the first task will establish a common vocabulary regarding model bias, robustness, reliability, and explainability. The project team and users will then discuss the perceived benefits and harms of using AI in SmartCHANGE. Next, educational sessions will be organised to illustrate (primarily from the end-user perspective) the XAI models and methods developed in the project. Mock-up interfaces will be designed, and mixed quantitative and qualitative research will be used to assess the potential of such solutions. These will include human-grounded task-based evaluation of XAI approaches, [38], interviews, observing how users interact with XAI tools while performing selected tasks, and understandings trade-offs in model performance and potential use or misuse of XAI solutions. Expert users will help us define more narrowly what feasible and actionable mean in counterfactual explanations of the developed risk models and how they can be translated into (personalised and contextualised) interventions leading to healthier lifestyles.

C. Application

The SmartCHANGE solution will be delivered via two distinct tools, able to communicate in a bi-directional manner: i) a web application for health professionals and ii) a mobile application for children, youth and their families.

- **Tool for health professionals** will support health professionals in making predictions of future risks for NCDs, particularly cardiovascular and metabolic diseases, which will be achieved by analyzing data on biological risk factors available in health records and supplemented with data on lifestyle behaviours (entered directly into the tool or provided via the tool for citizens). It will also allow for "what-if" modelling to estimate risk reduction probability ("what") through the application of specific interventions ("if"). Last but not least, the application will fit into the daily practice of the professionals, providing a simple overview of the risk factors and how they might address these. Even though elaborate analysis will be performed, a simplified version will be presented to the professionals providing them to use the application during a consult.
- **Tool for citizens** will work alongside the tool for health professionals and collect data on lifestyle behaviours using wearable activity monitors and other sources. The application will use visual analytics tailored to each user group and can be used independently or in combination for effective dialogue between citizens and health professionals. For instance, the family application will support parents and their children to reflect on their behaviour and provide realistic recommendations that can be applied in their daily lives. The key will be to engage both target groups to support reflection and behaviour change.

Both tools will offer evidence-based risk-lowering strategies defined with the guidance of health professionals and the support of other involved stakeholders. Selecting effective behaviour change strategies in different situations is a difficult task. Because different methods influence human behaviour on different mechanisms; for example, goal-setting on outcome may influence behaviour through motivation, and instruction on how to perform behaviour can affect work via targeting a belief system. The recently published tool on behaviour change and their mechanism of action in the Human behaviour change project, [39], can provide good guidelines for selecting appropriate strategies according to the targeted determinants. Additionally, they will prioritize risk factors, explain their impact, and suggest corresponding behaviour change strategies. However, precise quantification of the impact of specific behaviour changes on risk and comprehensive digital behaviour change interventions will be beyond the project's scope.

D. Evaluation through proof-of-concept studies

The feasibility of the SmartCHANGE tool for risk identification and primary prevention of cardiovascular disease (CVD) and metabolic disease will be tested in four real-world scenarios (Figure 4) across different European regions:

- In Portugal, we will collaborate with family physicians and paediatricians, recruiting families with children aged 6-10. The SmartCHANGE family app targets parents. Inclusion criteria: willing family physicians/paediatricians, children aged 6-10 under their care. Exclusion: simultaneous participation in another relevant study. Children will be conveniently recruited for gender balance. Goal: enrol 2-3 physicians/paediatricians, 50+ children (with parents) in experimental and control groups.

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- In Slovenia, four teams will be recruited from primary care centres, with an equal division between experimental and control groups. Both the professional tool and the family app will be utilized. The web-based tool will be aimed at healthcare professionals in existing multidisciplinary teams, which include paediatricians, nurses, and others, along with PE teachers from nearby schools. The SmartCHANGE family app will be focused on parents. To be eligible, a functional multidisciplinary team must exist at the community centre, and team members must register on the MYSLOfit app to gain access to SLOfit data. Additionally, there must be a minimum of 50 children aged 6-11 under the care of the core team. Exclusion criteria will concurrently encompass participation in conflicting studies, applying to professionals and patients. Healthcare professionals will be sampled conveniently to ensure gender balance, with a target of approximately 20 professionals across the four teams, involving over 100 adolescents under their care.
- In Finland, collaboration is being undertaken with school nurses. The experimental group comprises three nurses and 50 students (ages 11 to 14). The SmartCHANGE tool will be employed by school nurses during health checks to promote healthier habits, while the SmartCHANGE app will provide support to adolescents. A control group of the same size will adhere to standard procedures. Nurse recruitment will involve convenience sampling connected to national fitness evaluations (Move! measurements, [40]) and school health checks. Functional capacity is assessed by Move!, which supplements evaluations for 5th and 8th graders, with approximately 90,000 students participating annually. Subsequent health examinations are conducted by nurses, often involving parents. The objective is to enrol six nurses and over 100 students, ensuring an even distribution between the experimental and control groups.
- In the Netherlands, child public healthcare facilities linked to schools will be recruited. These facilities offer routine care to children aged 0-19 years. The tool for professionals will be utilised alongside the family application. Inclusion criteria for children are as follows: 1) age between 11-14 years; 2) attendance at schools affiliated with the participating child public health care facilities. The exclusion criterion will involve concurrent participation in another study that could impact the outcomes of both studies. This criterion will be applicable at both the healthcare professional and patient levels. Overall, enrollment of 2-3 child public health care facilities and approximately 50 children (and their parents) in the experimental group is anticipated. A control group of equal size will be recruited through the standard procedure.

Although the ultimate goal of the SmartCHANGE tool is to improve the overall health of youth, this is not easy to assess directly in this age group. In addition, given the time frame and

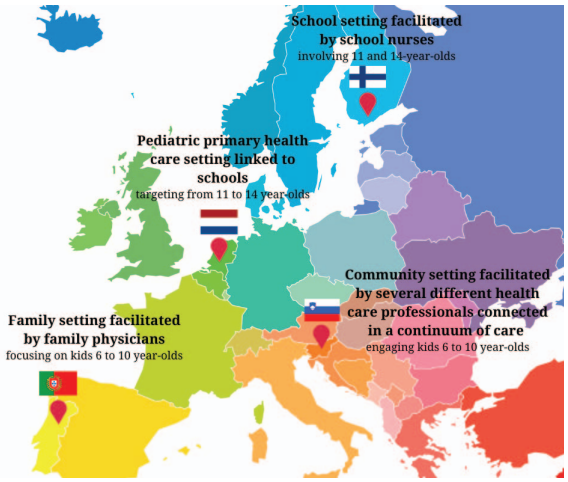


Fig. 4. Map of pilot studies.

the resources available for this project, it was not viable to plan a full-scale RCT that would provide robust evidence for the effectiveness of SmartCHANGE in improving health. Instead, we envisaged a non-randomised, controlled proof-of-concept study that will test the tool's implementation in real-world scenarios. We plan to document how much health professionals benefit from the tool in decision-making during daily clinical practice. As an intermediary outcome, we will examine behaviour change driven by the tool and guided by the health practitioners. Data collection will encompass feasibility outcomes, usability and user satisfaction, and clinical outcomes such as aggregated risk estimates, individual behavioural and biological risk factors for CVD, and other relevant health outcomes. Surveys and interviews will be conducted after the study period to evaluate the feasibility, usability, and explainability of the SmartCHANGE tools. User feedback will be analysed to ensure personalised risk-lowering strategies consider gender, diversity, cultural, and socio-economic differences. The study will also include limited efficacy testing, with the primary outcome being the change in aggregated risk produced by the SmartCHANGE tool. Intermediary outcomes will include changes in individual lifestyle behaviours (e.g., physical activity, sedentary time, sleep, diet, smoking, and excessive alcohol use). Secondary outcomes will involve changes in awareness of healthy lifestyles, health literacy, quality of life, and other relevant outcomes identified through participatory design involving health professionals and citizens. During one year, the SmartCHANGE tool will propose behaviour changes guided by specific healthcare professionals at each study site following the initial risk assessment. Data on biological risk factors from health records will be collected by SmartCHANGE and combined with lifestyle behaviour information. These inputs will fuel risk prediction models, generating future NCD risk estimates. A module will reveal individual risk factor contributions, aiding the prioritisation of preventative strategies. The tool will also support "what-if"

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scenarios, calculating risk reduction due to specific changes. It will suggest evidence-backed risk-lowering tactics based on priorities and aid health professionals in motivational interviews, goal setting, and overcoming barriers. Overall, SmartCHANGE empowers health pros to pinpoint high-risk children, emphasise key risk factors, and offer tailored risk-reduction advice, adapting interventions over time.

Clinical outcomes will be assessed at baseline, midterm (after six months), and after the end of the study (after 12 months). The intervention will be compared to usual care, tailored to different settings across the study sites.

IV. Conclusion

The SmartCHANGE project aims to address the significant public health burden of non-communicable chronic diseases (NCDs) in developed countries by developing trustworthy AI-driven decision-support tools. These tools will accurately assess risk factors in children and youth, promote optimised strategies to reduce the long-term risk of NCDs and fill the gap in current tools primarily focusing on adults. By targeting interventions towards children and adolescents, who are in crucial stages of establishing healthy habits, the project aims to prevent or delay the onset of NCDs later in life.

It is essential to acknowledge that the actual implementation and outcomes of the SmartCHANGE project may evolve and adapt as the research progresses. The complexities of the project, the availability of data, and the development of AI technologies may influence the specific methodologies and tools employed. Therefore, while the outline of the project provides a solid foundation, it is crucial to remain open to potential modifications and improvements as new insights and advancements emerge in NCD prevention and AI-driven decision support.

Through ongoing research and collaboration, the SmartCHANGE project will seek to contribute valuable insights and innovations to NCD prevention, ultimately benefiting the health and well-being of children and youth.

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